

Aditya Singh

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Summary

Software Engineering undergraduate with hands-on experience in Java, JavaScript, and geospatial data visualization. Skilled in developing responsive web applications and interactive 3D simulations using modern technologies. Strong foundation in object-oriented programming, data structures, and problem-solving. Passionate about building scalable, user-focused solutions and continuously improving through real-world projects and learning.

Skills

- **Languages:** Java | JavaScript | SQL | HTML5 | CSS3 |
- **Frontend Development:** Responsive Design | DOM Manipulation
- **Backend :** Spring Boot (*Learning*)
- **Database:** MySQL (Basic Queries, Data Handling)
- **Core Concepts:** Object-Oriented Programming (OOP) | Exception Handling | Multithreading | File Handling | Data Structures (Basic)
- **Libraries / Technologies:** Three.js | Geospatial Data Processing
- **Tools & Platforms:** Git (Version Control) | GitHub | VS Code | IntelliJ IDEA

Experience

Software Engineering Intern | ADRDE – DRDO, Agra

May – June 2025

- Contributed to development of **geospatial data processing systems** for defense-related applications
- Processed and analyzed structured datasets to improve **data accuracy, consistency, and usability**
- Assisted in designing and testing software modules within a **real-world R&D environment**
- Collaborated with cross-functional teams following **Software Development Life Cycle (SDLC)** practices
- Supported **technical documentation and system workflows** for better project maintainability
- Gained exposure to **scalable system design, debugging, and performance optimization techniques**

Projects

3D UAV Flight Path Visualizer

<https://github.com/Adityasingh076/3D-UAV-Flight-Path-Visualizer.git>

- Engineered an **interactive 3D simulation system to visualize UAV and satellite trajectories**
- Built a data pipeline to process **CSV-based geospatial datasets** (latitude, longitude, orientation)
- Applied coordinate transformation and computational geometry to map GPS data into 3D space
- Developed features like **dynamic camera tracking, trajectory rendering, and motion simulation**
- Optimized rendering performance for real-time visualization and smooth animations
- Structured code for **efficient data flow and scalable visualization logic**

Brew Tech Coffee House – E-Commerce Platform

<https://github.com/Adityasingh076/BREW-TECH.git>

- Developed a **responsive e-commerce web application** focusing on clean UI and seamless user experience
- Implemented **client-side cart system using localStorage** for persistent session management
- Built **dynamic product components** including modals and filtering system
- Designed and implemented **real-time billing and invoice generation module**
- Applied **DOM manipulation and event-driven programming** for interactive features
- Ensured **cross-device compatibility and performance optimization**

Smart Bottle Turbidity Visualizer | JavaScript, Data Visualization, Frontend Engineering

<https://github.com/Adityasingh076/Smart-Bottle-Turbidity-Visualizer.git>

- Designed and developed a **data visualization system** to monitor and interpret water turbidity levels
- Implemented **dynamic data handling using JavaScript** for real-time updates and processing
- Built **interactive UI components** to represent sensor-based data clearly
- Applied **data mapping techniques** to convert raw inputs into meaningful visual insights
- Optimized rendering logic for **smooth performance and user interaction**
- Followed **modular code structure** for scalability and maintainability

Education

- **Bachelor of Technology (B.Tech)**

Eshan College of Engineering, Mathura | 2022 – 2026

CGPA: 8.4 / 10 (till 7th Semester)

- **Class XII – 68% | 2021 – 2022**

- **Class X – 60% | 2019 – 2020**

Shivalik Cambridge School, Agra (CBSE)